

Boundary back-propagation verification: complete affected proof bodies

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This verification reader includes complete revised earlier chapters and the complete authoritative boundary proofs. Historical originals and all exact edits are retained beside this source. References to other chapters retain their identifiers for reintegration into the full book.

Result R51: the constructed Deligne comparison maps and constituent coupling.

The detailed Deligne sidebar proves the image, kernel and cone maps for the same compact-to-ordinary comparison, and the complementary degree duality with its actual boundary line. For every specified invariant $F \subseteq E$ and divisor depth $m \geq 1$, its original curve has the logarithmic complex

$$\mathcal{K}_{F,m} = [F\mathcal{O} + E\mathcal{O}(-m\infty) \xrightarrow{d} (F\mathcal{O} + E\mathcal{O}(-m\infty)) \otimes \Omega^1(\log \infty)].$$

Its complete cohomology is F in degree zero, $(E/F) \otimes \mathbb{T}$ in degree two and zero in degree one, with $\mathbb{T} = H^1(\Omega^1)$ and the retained trace $\text{tr}[dz/z] = 1$. The higher-depth quotient contracts by division by the displayed nonzero integer coefficients; its dual retains all meromorphic powers and the twist by $\mathcal{O}(-\infty)$. Each split lift distinguishes the represented kernel from external absence through their disjoint inclusions into the full kernel carrier.

The same original metric, $G(t) = G_j + t(G_i - G_j)$, gives the invariant quotient metric and coupling map proved in (DS53)–(DS65). With those specified maps, its endpoint identity is

$$\log(V_i/V_j) = \log \frac{\det G_F(1)}{\det G_F(0)} + \log \frac{\det Q_+}{\det Q_0} - \sum_{r=1}^{\infty} \frac{\text{Tr}(\Xi_F^r)}{r}.$$

The series has nonnegative terms before subtraction, and the sidebar proves its finite remainder. It also proves the exact original theta source defect $dK(z)H_t + r(z)Ia_t$, so the theta boundary and the retained constituent pass through their specified successive quotients. The two positive determinants can use R50 on their respective compressed matrices; the coupling subtraction remains in the same upper estimate.

Result R52: a degree-one exponential family with exact periods. For the original cyclic polynomial χ and its specified primitive Φ , the delivered exponential note proves

$$\mathcal{H} = \text{coker}(u\partial_S + \chi(S) - t), \quad \mathcal{H} \cong \mathbb{C}[u, t]^q, \quad \mathcal{H}/(u, t)\mathcal{H} \cong E, \quad (-u\nabla_t) \bmod (u, t) = A.$$

Its coefficient-module division, original theta observation, finite-rank source deformation, infinity chart and dual residue pairing are all retained with complete proofs in the exponential note and sidebar. The new determinant proof (XD1)–(XD26) computes the entire period determinant for the stated oriented rays:

$$\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp \left(\frac{\text{Tr} \Phi_t(A(t))}{u} \right).$$

All Gamma factors and monomial phases are given in (XD16)–(XD18). The proof uses the actual twisted-reduction matrices, keeps their off-diagonal corrections, and proves that their trace corrections vanish. It establishes nonvanishing across every repeated-root parameter. On the original canonical metric it gives

$$\log \det B_N^{\text{per}} = \log C_q(u) + 2 \operatorname{Re} \frac{\operatorname{Tr} \Phi_t(A(t))}{u} - \log V_N.$$

Subtracting at the same parameters retains the exact original endpoint cost. The translation companion (DT1)–(DT13) further proves the complete maps under $\chi_a(S) = \chi(S - a)$: the coefficient action shifts by aI , the period gains the scalar $\exp[(-\Phi(-a) - ta)/u]$ and the full Pascal matrix, and the finite-field realization retains the character sheaf $\mathcal{L}_\psi((-\Phi(-a) - ta)/u)$ on the original (u, t) -base; its point restriction is the stated constant character line. The complete derived direct-image map and its projections are proved in (DT.14)–(DT.16). The original arithmetic fibre remains $a = 0$. These explicit maps preserve the canonical metric comparison and the full inherited endpoint inequalities.

The full connection in the same cyclic coefficient frame is now (XD.u1)–(XD.u7). Define its columns by the literal monic division

$$\Phi_t(S)S^b = (\chi(S) - t)Q_b^\Phi + R_b^\Phi, \quad (C_\chi(t))_b = R_b^\Phi, \quad (B_\chi)_b = (Q_b^\Phi)'. \quad (XD.u1)$$

The differential remainder is $R_b^\Phi - u(Q_b^\Phi)'$. With the original branch and all oriented contours retained,

$$\Omega_u = -C_\chi(t)/u^2 + B_\chi/u, \quad \Pi_u = \Pi\Omega_u, \quad \Omega_t = -A(t)/u, \quad \Pi_t = \Pi\Omega_t, \quad \partial_u\Omega_t - \partial_t\Omega_u + [\Omega_u, \Omega_t] = 0. \quad (XD.u2)$$

The degree-zero connection is $\partial_u - \Phi_t/u^2 + 1/u$; the degree-one connection is $\partial_u - \Phi_t/u^2$. Their cochain identity, complete division, and locally uniform contour differentiation are proved in (XD.u1)–(XD.u6). Here

$$C_\chi(t) = \Phi_t(A(t)), \quad (B_\chi)_{bb} = (b+1)/(q+1), \quad \partial_t B_\chi = 0, \quad \operatorname{Tr} B_\chi = q/2. \quad (XD.u3)$$

Thus the preceding determinant retains its full matrix connection, with the same original constant

$$K_q = (q+1)^{-q/2} e^{\pi i q/2} \det W \prod_{b=0}^{q-1} \Gamma((b+1)/(q+1)), \quad W_{jb} = e^{2\pi i j(b+1)/(q+1)} - 1. \quad (XD.u4)$$

On the exact primary slice $\chi = h = (s-\rho)^m$, $t = 0$, set $n = m+1$, $c = -(-\rho)^n/n$, $\beta_b = (b+1)/n$ and $d_b = n^{\beta_b-1} e^{\pi i \beta_b} \Gamma(\beta_b)$. The original Pascal matrix has $P_{ab} = \binom{b}{a} \rho^{b-a}$ for $a \leq b$, and the ordered period matrix is exactly $\Pi_s = e^{c/u} W D(u) P$, where $D(u) = \operatorname{diag}(d_b u^{\beta_b})$. For an original p -dimensional ideal inclusion $I = J_{m-p} C_F$, whose J_{m-p} columns are $(s-\rho)^{m-p}, \dots, (s-\rho)^{m-1}$ in the unchanged s -basis, (SC.u5)–(SC.u7) compute

$$\det(I^* \Pi_s^* \Pi_s I) = |\det C_F|^2 n^p (p+1) e^{2p \operatorname{Re}(c/u)} \prod_{b=m-p}^{m-1} |d_b|^2 |u|^{2\beta_b}. \quad (XD.u5)$$

The full Gram is $e^{2 \operatorname{Re}(c/u)} C_F^* D_{m-p}^* (W_{m-p}^* W_{m-p}) D_{m-p} C_F$, with $W_{m-p}^* W_{m-p} = n(I_p + \mathbf{1}\mathbf{1}^*)$. Its comparison with the original theta Gram divides the determinant by $\det(I^* G_N I)$, and uses $(I^* G_N I)^{-1}$ on that complete matrix. The proof retains the original source unit and every frame determinant. Its image plane is constant in u , so its normal u -map and u -curvature vanish on precisely this primary $t = 0$ slice. The general two-parameter normal map remains $N \Pi \Omega_a I$ for $a = u, t$, as in (SC.u1)–(SC.u4).

For AP's original packet polynomial h , full fixed Taylor representative ν , and fixed poles $p \in P_\nu$, the localized complex now has the entire connection (APU1)–(APU23). Put $f = \Phi_h - ts$ and

$$Q_p = \frac{f(s, t) - f(p, t)}{s - p} - \frac{h(s) - t}{\deg h + 1}.$$

Its ordinary quotient is the constant $1/(\deg h + 1)$, so its derivative correction is exactly zero. In AP6's original polynomial and simple-pole columns,

$$\Omega_u^{\text{loc}} = \begin{bmatrix} \Omega_u^{\text{pol}} & -Q/u^2 \\ 0 & -\text{diag}(f(p, t))/u^2 \end{bmatrix}.$$

The weighted-residue matrix is $\mathcal{R} = [0, W_{\text{res}}]$, with $W_{\text{res}} = \text{diag}(e^{f(p, t)/u})$; the positively oriented loop map is exactly $(2\pi i)\mathcal{R}$. Both maps are horizontal in u, t . The same complete ν intertwines these connections on the transported gauge frame, with target weight w/ν . The original endpoint-source map is $\sigma_h[\nu]J_0$, with every inverse Taylor coefficient retained, as proved in (APU18)–(APU22). Finally the original entire transport $C_a = \Pi_{\text{loc}}(u, t)^{-1}\Pi_{\text{loc}}(u, t + ua)$ has the exact derivative

$$(C_a)_u = -\Omega_u^{\text{loc}}(u, t)C_a + C_a(\Omega_u^{\text{loc}}(u, t + ua) + a\Omega_t^{\text{loc}}(u, t + ua)).$$

Its t -derivative and the differentiated weighted-residue and loop diagrams are proved in (APU23). The extra $a\Omega_t^{\text{loc}}$ is the derivative of the original parameter shift. The identification $h = \chi_{h,1}$ is the literal variable isomorphism (PC4); at higher tensor degree the cyclic and product families retain their specified η_k and source maps rather than being assigned a common coefficient rank.

Result R56: the actual critical-comparison kernel through periods and tensors. The full proof (PC1)–(PC41) identifies the single-packet polynomial $\chi_{h,1} = h$ by its explicit variable isomorphism $\iota_1 : [P(S)] \mapsto [P(s)]$. The original arithmetic inclusion is $\eta_h = M_{v_h}\iota_1$, with $v_h = j_h(2\xi/h)$ and every inverse jet coefficient retained. It proves

$$J_h r_N = \eta_h, \quad j_E = \eta_h^{-1} J_h, \quad j_E r_N = I, \quad q r_N = \sigma_h \eta_h.$$

The critical comparison is consequently the actual weighted map $\mathcal{C}_h = \Gamma\sigma_h\eta_h$; its kernel is the full off-critical primary summand. Each surviving critical jet is recovered with the original $i^{-j}/j!$ factor and the inverse of the displayed local arithmetic unit.

For every original integer tensor degree $k \geq 1$, retain $E_k = \mathbb{C}[S]/(\chi_{h,k})$, its actual weighted inclusion η_k , and the factorwise completed-projective comparison

$$\mathcal{C}_k = \Gamma^{\widehat{\otimes}^k} \sigma_h^{\widehat{\otimes}^k} \eta_k.$$

Let $\chi_{c,k}$ be the monic sum polynomial obtained from the all-critical tuples, with the complete maximum nilpotent length at each literally equal sum; for no critical factors set it to 1. The full proof constructs the variable isomorphism on the tensor carriers, preserves the original unit, and gives a continuous jet retraction proving that the finite image survives completion. It yields

$$\boxed{\ker \mathcal{C}_k = (\chi_{c,k})/(\chi_{h,k}).}$$

At a sum λ with full multiplicity m_λ and critical-tuple multiplicity r_λ , its exact local kernel is the image of $[b] \mapsto [x^{r_\lambda} a_\lambda(x)b]$ from $\mathbb{C}[x]/(x^{m_\lambda - r_\lambda})$, where $x = S - \lambda$ and the local unit in $\chi_{c,k}$ is retained. This includes the additional jets when different tuples share a sum.

For the period matrix of this same original polynomial, $\mathcal{C}_{k,t} = \mathcal{C}_k \Pi_k^{-1}$ has kernel $\Pi_k \ker \mathcal{C}_k$. Its exact sum-action defect is

$$\mathcal{C}_{k,t} A_{k,t}^{\text{per}} - M^{[k]} \mathcal{C}_{k,t} = t \mathcal{C}_k(e_{0,k}) \ell_k \Pi_k^{-1}.$$

For a proper nonzero kernel it has the proved rank-one restriction, with kernel $\Pi_k(\ker \mathcal{C}_k \cap \ker \ell_k)$. The curvature and neighboring-minor identities in R54 therefore measure this actual critical-comparison kernel. The original theta metric is connected by the complete congruence and derivative cancellation. All-critical and no-critical packets retain their different empty and full kernels, even though both endpoint curvatures vanish.

The same weighted observer now has its full singular-parameter equation (PC.u1)–(PC.u3):

$$\mathcal{C}_{k,u,t} = \mathcal{C}_k \Pi_\chi(u, t)^{-1}, \quad \partial_u \mathcal{C}_{k,u,t} = \mathcal{C}_k (C_\chi(t)/u^2 - B_\chi/u) \Pi_\chi^{-1}, \quad \ker \mathcal{C}_{k,u,t} = \Pi_\chi \ker \mathcal{C}_k.$$

Here the polynomial is literally $\chi = \chi_{h,k}$; the product-period restriction on $E_h^{\otimes k}$ is instead the actual rectangular form $\eta_k^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_k$. The source identity and parameter cyclic relation following (XD.u7) are the exact maps between those two domains. For the original kernel inclusion I_k , its complete normal maps are $Z_a = N \Pi_\chi \Omega_a I_k$, $a = u, t$, and all four curvature coefficients are

$$\partial_a \partial_b \log \det H = \text{Tr}(H^{-1} Z_a^* Z_b).$$

At $t = 0$, the order-two coefficient preserves this ideal, so $Z_u = u^{-1} N \Pi_\chi B_\chi I_k$, $Z_t = 0$. These are calculated matrix entries on the same kernel; neither the full ideal nor its nilpotent jets have been suppressed.

The actual tensor coefficient map is also now retained in (PC30a)–(PC30g). On an ordered local tuple let $x_i = s_i - \rho_i$, $\lambda = \sum_i \rho_i$, $D = \sum_i (m_i - 1)$, and $U_\rho = \prod_i j_{\rho_i}(v_h)$. Over the specified coefficient localization $\mathcal{R} = A_{\ker \alpha}$, for an actual map to characteristic $p > \max_i m_i$, the exact map and its saturation basis are

$$V_j = \sum_{\substack{0 \leq b_i < m_i \\ |\mathbf{b}|=j}} \frac{x^{\mathbf{b}}}{\prod_i b_i!}, \quad W_j = U_\rho V_j, \quad \eta_\rho((S - \lambda)^j) = j! W_j, \quad \Lambda_U = \bigoplus_{j=0}^D \mathcal{R} W_j.$$

Its full containing tensor module retains the explicitly split free complement, and

$$\Lambda_U / \eta_\rho \mathcal{C}_{\mathcal{R}} \simeq \bigoplus_{j=0}^D \mathcal{R} / (p^{v_p(j!)}), \quad \ker \bar{\eta}_\rho = ((S - \bar{\lambda})^p), \quad \text{im } \bar{\eta}_\rho = \bigoplus_{j=0}^{\min(D, p-1)} B \bar{W}_j.$$

Here B is the specified target k_0 , or the explicit quotient $\mathcal{R}/p\mathcal{R}$. The kernel is zero when $D < p$. The original source index is $D+1$, while the index on the actual specialized image is $\min(p, D+1)$; the full free resolution and Tor connecting map in (PC30e) prove their exact relation. When the specified map includes all one-factor CRT data and $p > m_\rho$ for every retained primary order, (PC30f)–(PC30g) also retain every new residual collision:

$$\psi_{h,k}(X) = \prod_{\mu} (X - \mu)^{\max_{\alpha}(\sum_i \rho_i) = \mu \min(p, 1 + \sum_i (m_i - 1))}, \quad \ker \bar{\eta}_k = (\psi_{h,k}) / (\bar{\chi}_{h,k}), \quad \text{im } \bar{\eta}_k \simeq k_0[X] / (\psi_{h,k}).$$

The product is over attained residual sums; if the carrier is empty, its product is 1 and both modules are zero. The last arrow is the proved module map $[P] \mapsto \bar{U}_k P(\bar{S})$; its unweighted counterpart is the corresponding algebra map.

For the complete primary packet at $t_i = 0$, the ordered product boundary is explicitly calculated in (SPBT.1)–(SPBT.3), (BPC.1)–(BPC.14), (BTP.1)–(BTP.11), and (BTF.1)–(BTF.11). Writing $n = m + 1$, its ordinary complex invariant dimension is

$$d_{0,k} = \frac{m^k + m(-1)^k}{m+1}.$$

With E_0 selecting $\sum_i (b_i + 1) \equiv 0 \pmod{n}$, the exact coefficient, source and period projectors satisfy

$$Q_0 = P_k^{-1} E_0 P_k, \quad \hat{Q}_0 = U_k Q_0 U_k^{-1}, \quad \mathcal{P}_0 \Pi_{\text{prod}} U_k^{-1} = \Pi_{\text{prod}} U_k^{-1} \hat{Q}_0.$$

The terminal original monomial survives precisely when $n \mid k$, and retains its arithmetic eigenvalue $a^{k\rho}$. The exact compression and its outgoing nilpotent map are (BTP.7)–(BTP.8). On the actual phase specialization with $p > m + 1$, the common additive factor remains $\mathcal{L}_\psi(kc/u)$: the invariant dimension is $d_{0,k}$ when $kc = 0$ and zero otherwise, and the Swan conductor is respectively zero

or m^k . Every surviving boundary orbit retains its full product of negative Gauss constants and weight k . The common cyclotomic projector maps in (BTF.6) give their exact two base changes; they do not assign a characteristic- p coefficient scalar to a characteristic-zero Gauss line. The amplified coefficient still contains its original $K!$, $K = k(m-1)$, with image zero exactly when $K \geq p$ on a coefficient stage retaining the inverse unit. The case $m = 1$ has $K = 0$ and no factorial loss. Moreover, in the retained off-critical quartet CRT corner for $k \geq p$, (BTF.9) proves

$$1 = p(\rho + \bar{\rho} - 1)\mathcal{D}, \quad p^{-1} = (\rho + \bar{\rho} - 1)\mathcal{D}$$

from the actual separating idempotent's divided difference. That larger stage has no unital characteristic- p map; the original local coefficient map, its precise saturation, kernel, Tor class and character remain as displayed above.

Result R57: computed Laplacian control and its curvature comparison. The full original-metric calculation (LC1)–(LC16) uses $A_t = A + tR$, $R = e_0\ell$, and $W_t = A_t^*G_N + G_NA_t - kG_N$. For $q > 1$, $\ell e_0 = 0$, so $R^2 = 0$. The two actual G_N -orthogonal columns $e_0, G_N^{-1}\ell^*$ give the exact additional control spectrum

$$\text{spec}_{\neq 0}(G_N^{-1}(W_t - W_0)) = \{|t|\sigma_N, -|t|\sigma_N\}, \quad \sigma_N^2 = (e_0^*G_N e_0)(\ell G_N^{-1}\ell^*),$$

when $t \neq 0$. For $q = 1$ the same difference is exactly $2\text{Re } t$ times identity. For the retained reflection-stable packet, the complete earlier source-minor calculation evaluates $\epsilon_N^2 = \alpha_{N+1}\Delta_N/\det K_N$, so the actual quantity

$$E_N(t) = \epsilon_N + \begin{cases} |t|\sigma_N, & q > 1, \\ 2|\text{Re } t|, & q = 1 \end{cases}$$

bounds the control on the same metric. With this computed value, every point z of the original-metric numerical range of $A_t^2 - kA_t$ satisfies

$$\text{Re } z \leq \frac{E_N(t)^2 - k^2}{4}, \quad (\text{Im } z)^2 \leq E_N(t)^2 \left(\frac{E_N(t)^2 - k^2}{4} - \text{Re } z \right).$$

The proof retains every original input vector in homogeneous coordinates and transports the complete inequality through Π with metric $\Pi^{-*}G_N\Pi^{-1}$.

The same source factors also control the calculated constituent curvature. For its actual G_N -projection P_F , set $a_\perp = \|(I - P_F)e_0\|_{G_N}^2$ and $b_F = \|P_F G_N^{-1}\ell^*\|_{G_N}^2$. Let $m(t), M(t)$ be the actual least and greatest eigenvalues of $G_N^{-1}\Pi^*\Pi$. Its positive curvature coefficient obeys

$$\frac{m(t)}{M(t)} \frac{a_\perp b_F}{|u|^2} \leq c_F(t) \leq \frac{M(t)}{m(t)} \frac{a_\perp b_F}{|u|^2} \leq \frac{M(t)}{m(t)} \frac{\sigma_N^2}{|u|^2}.$$

The complete proof supplies the exact nonnegative projection losses between $a_\perp b_F$ and σ_N^2 . These statements retain the actual period-comparison eigenvalues and original theta masses; no uniform packet-size bound is inserted.

The same numerical-range calculation is now transported in both original parameters by (LC.u1), with the unchanged $E_N(t)$ and G_N :

$$\begin{aligned} \partial_u \tilde{G}_N &= -\Pi^{-*}G_N\Omega_u\Pi^{-1}, & \partial_{\bar{u}} \tilde{G}_N &= -\Pi^{-*}\Omega_u^*G_N\Pi^{-1}, \\ \partial_u \tilde{L}_t &= \Pi[\Omega_u, L_t]\Pi^{-1}, & \partial_u \tilde{A}_t &= \Pi[\Omega_u, A_t]\Pi^{-1}. \end{aligned}$$

Here $\Omega_u = -C_\chi(t)/u^2 + B_\chi/u$ is the full original cyclic connection. The exact original source commutator retains $r_N[A, \Omega_u]j_E + dK_N^{\text{prim}}\Omega_u j_E$, as proved in (DS74) and carried through (LC.u1). Thus the preceding original-metric parabola holds for every retained $u \neq 0$, with its same control

value and the displayed moving metric, rather than an uncomputed target norm. The comparison numbers in (LC14)–(LC16) are now the actual $m(u, t), M(u, t)$ of $G_N^{-1}\Pi(u, t)^*\Pi(u, t)$. For each constituent and each neighboring inclusion, (SC.u1)–(SC.u4) give the complete Hermitian Hessian $\text{Tr}(H^{-1}Z_a^*Z_b)$; at nonzero t , both terms of Ω_u and their cross pairing remain. On the precise single-primary ideal slice the full matrix and ideal determinant are evaluated by (SC.u5)–(SC.u7), including the original comparison Gram.

Result R59: the joint critical-jet and circle observation. The full density proof obtains the original exponential tails from the gamma ray integral and the alternating Dirichlet series, including every canceled value $v_h(\rho) = g^{(m)}(\rho)/h^{(m)}(\rho) \neq 0$. It proves density on the required analytic strip, with its Cauchy–Schwarz half-width retained, and computes the completed circle quotient exactly. If $b = p_L$ is the squarefree product of the sampled roots of $\chi = \chi_{h,k}$, then

$$\ker \theta_L = (b)/(\chi), \quad G_N(L) \downarrow G_\infty(L), \quad \text{rad } G_\infty(L) = (b)/(\chi).$$

For $k \geq 2$ the complete difference $G_N(L) - G_\infty(L)$ is positive definite at every finite admitted degree: a nonzero minimizing polynomial cannot vanish on the infinite complement of the selected sample atoms. The limiting kernel retains the full unsampled primary blocks and all nonconstant jets at sampled roots.

The proof (DC1)–(DC30) compares this with the actual weighted tensor-critical map C_k , whose kernel is $(a)/(\chi)$ for $a = \chi_{c,k}$. Put $d = \gcd(a, b)$ and $l = \text{lcm}(a, b)$. Using the complete critical inverse jets and the original weighted circle coordinates gives

$$\text{im}(C_k, \theta_L) \simeq \text{im } C_k \times_{\mathbb{C}[S]/(d)} \text{im } \theta_L \simeq \mathbb{C}[S]/(l).$$

The joint kernel is $(l)/(\chi)$ and the common quotient is $\mathbb{C}[S]/(d)$. At each original sum root, their quotient lengths are respectively the maximum and minimum of the critical cyclic length and the sampled length in $\{0, 1\}$; all local unit cofactors are retained in the kernel inclusions. A factorization $\theta_L = TC_k$ exists exactly when $b \mid a$, with its explicit inverse-unit and reduction formula. The reverse factorization exists exactly when $a \mid b$. At $k = 1$, full-order cancellation proves that every sampled selected root has positive weight and $b \mid a$. For general tensor degree, the extra circle summand has defining polynomial b/d and the explicit original polynomial source lift in (DC29). Its existence criterion asserts no existence of an off-critical zero.

All four observations transport through the same original period matrix. For the role $s \in \{a, b, d, l\}$, retain its actual map T_s and multiplier M_s . Then

$$T_{s,t} = T_s \Pi^{-1}, \quad u \partial_t T_{s,t} - M_s T_{s,t} = t T_s(e_0) \ell \Pi^{-1}.$$

Their kernels, intersection, sum, crossed observations and complete rank-one action defects are calculated in (DC23)–(DC30). Their Split-Zero lifts retain each support and its receiving supported zero.

The entire same diamond now has its exact u -equation (DC31), for all four formal roles $s \in \{a, b, d, l\}$:

$$T_{s,u,t} = T_s \Pi(u, t)^{-1}, \quad \partial_u T_{s,u,t} = T_s (C_\chi(t)/u^2 - B_\chi/u) \Pi(u, t)^{-1}.$$

Its kernels are ΠK_s , and its specified right inverses, intersection and sum diagrams are the same ones in (DC23)–(DC30). Multiplication by the original common quotient maps proves their differential compatibility, retaining all four roles and every inverse-unit factor. For the actual circle weights put $D_\nu = \text{diag}(\nu_n)_{n \in \mathbb{Z}_L}$. The exact observed form and its derivatives are (DC32):

$$\begin{aligned} H_b^{\text{obs}} &= \Pi^{-*} T_b^* D_\nu T_b \Pi^{-1}, & \ker H_b^{\text{obs}} &= \Pi K_b, \\ \partial_u H_b^{\text{obs}} &= -\Pi^{-*} T_b^* D_\nu T_b \Omega_u \Pi^{-1}, & \partial_{\bar{u}} H_b^{\text{obs}} &= -\Pi^{-*} \Omega_u^* T_b^* D_\nu T_b \Pi^{-1}. \end{aligned}$$

Every retained atom has its original positive weight; an empty sample set gives the empty observation and its full kernel. The shared and additional circle sums, polynomial source lift and receiving support labels therefore retain their original maps while the parameter varies.

Result R62: quotient curvature and the complete fourth derivative. Retain the exact original coefficient sequence $0 \rightarrow F \xrightarrow{I} E \rightarrow E/F \rightarrow 0$ and a constant quotient lift J_F . Its least-period-norm lift and quotient Gram $K(t)$ are the full Schur formulas (FC21). With $C = [I, J_F]$ they satisfy

$$\det(I^*\Pi^*\Pi I) \det K = |\det C|^2 |\det \Pi|^2, \quad \partial_{\bar{t}} \partial_t \log \det K = -\frac{|t|^2}{|u|^2} \mathfrak{c}_F(\Pi^*\Pi).$$

The positive constituent curvature is therefore compensated in this same coefficient quotient, with both curvatures vanishing at zero and with opposite nonzero leading coefficients. The full proof includes quotient-lift independence, basis covariance and the literal comparison with the original theta quotient metric.

The residue continuation evaluates the pure jets as well. If $\psi = \log \det(I^*\Pi(t)^*\Pi(t)I)$ and $\phi(t) = \log \det(I^*\Pi(0)^*\Pi(t)I)$ near zero, then

$$\left. \frac{d^4}{dx^4} \psi(x) \right|_0 = 2 \operatorname{Re} \phi^{(4)}(0) + 6 \mathfrak{c}_F(\Pi(0)^*\Pi(0))/|u|^2.$$

The full ordered recursion for $\Pi^{(n)} = \Pi Q_n$, its explicit Q_4 including $3R^2/u^2$, and the complete trace-log formula in (RCX30)–(RCX32) evaluate the remaining pure term. The forced period equation retains $u\Pi R$, and its original source primitive retains tKR . The exact Pascal translation preserves the t -derivatives of order at least two at fixed u , while its scalar factor retains the first-derivative trace shift $2p \operatorname{Re} a$. Its pure u -derivatives are the explicitly retained terms in (RCX.u5)–(RCX.u6) below.

The arithmetic norm theorem, sharp endpoint products and upper comparison therefore enter the existing Toda, Gamma and exterior maps with complete proof texts. The finite upper determinants in R45 are the concrete next quantities to estimate at the amplified degree; the explicit scalar estimate and the lower limit 4 currently coexist with the gap proved above.

The same constant quotient lift and the complete two-parameter period matrix give the exact compensating Hessian

$$\partial_{\bar{a}} \partial_b \log \det K = -\operatorname{Tr}(H^{-1} Z_a^* Z_b), \quad a, b \in \{u, t\}, \quad H = I^*\Pi^*\Pi I, \quad Z_a = N\Pi\Omega_a I.$$

Indeed the displayed Schur determinant identity holds at each (u, t) , its constant frame factor has zero derivative, and $\log |\det \Pi|^2$ is pluriharmonic where Π is invertible. Applying (SC.u2) proves the equality for both diagonal and both mixed entries, with the original quotient metric and frame preserved. The earlier quadratic vanishing statement is its t entry at $t = 0$; the u entry retains the B_χ/u normal term of (SC.u3).

The complete ordered derivative recursion now is (RCX.u1)–(RCX.u2):

$$Q_{0,0} = I, \quad Q_{r+1,s} = \partial_u Q_{r,s} + \Omega_u Q_{r,s}, \quad Q_{r,s+1} = \partial_t Q_{r,s} + \Omega_t Q_{r,s}, \quad \partial_u^r \partial_t^s \Pi = \Pi Q_{r,s}.$$

Flatness proves compatibility of all derivative orders. In particular,

$$\begin{aligned} Q_{2,0} &= C_\chi^2/u^4 + (2C_\chi - C_\chi B_\chi - B_\chi C_\chi)/u^3 + (B_\chi^2 - B_\chi)/u^2, \\ Q_{1,1} &= C_\chi A_t/u^3 + (A_t - B_\chi A_t)/u^2. \end{aligned}$$

All earlier pure t jets, including $3R^2/u^2$ in the fourth derivative, are the same $r = 0$ slice. Along a specified curve $(u(s), t(s))$, the exact metric variation inserted into the original residue Rayleigh-quotient formula is

$$\dot{H}_{\text{full}} = \Omega_{\text{path}}^* H_{\text{full}} + H_{\text{full}} \Omega_{\text{path}}, \quad \Omega_{\text{path}} = \dot{u} \Omega_u + \dot{t} \Omega_t, \quad H_{\text{full}} = \Pi^* \Pi.$$

The full forced-equation descendant also remains, by (RCX.u4):

$$2u\Pi_{tt} + u^2\Pi_{utt} + k\Pi_t + ku\Pi_{ut} + \Pi R + u\Pi_u R = \Pi\Omega_u L_t.$$

It composes with the unchanged theta-source map j_E and retains its original boundary primitive.

Finally, with $p = \dim F$ and $h_a(t) = -\Phi(-a) - ta$, the original translated phase gives

$$\psi_a - \psi = 2p \operatorname{Re}(h_a(t)/u), \quad \partial_u(\psi_a - \psi) = -ph_a(t)/u^2, \quad \partial_u^2(\psi_a - \psi) = 2ph_a(t)/u^3, \quad \partial_u \partial_t(\psi_a - \psi) = pa/u^2.$$

These are the exact derivatives proved in (RCX.u5)–(RCX.u6), through the full translated connection (DT.14)–(DT.16). Their Hermitian mixed derivatives vanish because the retained scalar difference is pluriharmonic, while these pure derivatives remain in the ordered recursion.

Result R68: the arithmetic cyclic image inside the full period family. For $d = \deg h$, the original one-factor exponential family over $\mathbb{C}[u, t]$ is $[\mathbb{C}[u, t][s] \xrightarrow{u\partial_s + h(s) - t} \mathbb{C}[u, t][s] ds]$. Use the original primitive $\Phi'_h = h$, its fixed additive constant, and the oriented rapid-decay contours in the admitted sector of $u \neq 0$. Their full matrix $\Pi_h(u, t)$ has columns $\int_{\Gamma_j} s^a e^{(\Phi_h(s) - ts)/u} ds$, $0 \leq a < d$. The complete period proof retains its determinant, ray phases, contour orientations and all coefficients. With independent parameters $t_1, \dots, t_k$, the unchanged rectangular matrix

$$\mathcal{P}(u, \mathbf{t}) = \left(\bigotimes_{i=1}^k \Pi_h(u, t_i) \right) \eta_{h,k}$$

is injective, because each period factor is invertible and $\eta_{h,k}$ is injective. The exact derivative at the undeformed parameter is

$$-u \sum_{i=1}^k \partial_{t_i} \mathcal{P}(u, \mathbf{0}) = \mathcal{P}(u, \mathbf{0}) A_{h,k}.$$

Differentiating each exponential inserts s_i ; the cyclic intertwining and commutation of the full tensor unit with $\sum A_i$ prove the formula. It uses the original sum coordinate, with no division by k . The full (DP23)–(DP33) proof supplies the coefficient-ring reductions, exact left inverse on this image, original metric pullback and separately defined finite-field external product. All nonsemisimple jets and root collisions have their stated domains there.

The complete singular-parameter connection acts on this same rectangular image. Write $f_i(s_i) = \Phi_h(s_i) - t_i s_i$ and retain the literal monic divisions

$$f_i(s_i) s_i^b = (h(s_i) - t_i) q_{i,b} + r_{i,b}, \quad (C_h(t_i))_b = r_{i,b}, \quad (B_h)_b = q'_{i,b}.$$

The differential remainder is $r_{i,b} - u q'_{i,b}$, with $C_h(t_i) = f_i(A_h(t_i))$. The original one-factor matrix is $\Omega_{u,h}(u, t_i) = -C_h(t_i)/u^2 + B_h/u$, and $\partial_u \Pi_h = \Pi_h \Omega_{u,h}$. The complete division and contour proof is in the boundary connection chapter and (XD.u1)–(XD.u6), applied to the displayed original h -variable. On the ordered product complex the degree- j cochain operator is exactly

$$\nabla_u^{(j)} = \partial_u - \frac{\sum_i f_i}{u^2} + \frac{k-j}{u}.$$

Its cochain identity follows one slot at a time from the degree-zero/degree-one identity: increasing exterior degree removes exactly one $1/u$ term. Every exterior sign and the original ordered form are retained.

Set $\Pi_{\text{prod}} = \bigotimes_i \Pi_h(u, t_i)$; let $\Omega_{u,\text{prod}}$ be the sum of the k matrices $\Omega_{u,h}(u, t_i)$ acting in their respective tensor slots. The product rule gives the exact original map

$$\partial_u \mathcal{P} = \Pi_{\text{prod}} \Omega_{u,\text{prod}} \eta_{h,k}.$$

For the full original positive tensor coefficient Gram G at the specified finite source cutoff, retain the explicit left inverse

$$\ell_\eta = (\eta_{h,k}^* G \eta_{h,k})^{-1} \eta_{h,k}^* G, \quad \ell_\eta \eta_{h,k} = I.$$

The inverse exists because $\eta_{h,k}$ is injective and G is positive on the original full tensor source. The complete parameter defect is therefore

$$\partial_u \mathcal{P} - \mathcal{P}(\ell_\eta \Omega_{u,\text{prod}} \eta_{h,k}) = \Pi_{\text{prod}}(I - \eta_{h,k} \ell_\eta) \Omega_{u,\text{prod}} \eta_{h,k}.$$

Substitution and factoring prove this equality, retaining the actual transverse component and full unit in $\eta_{h,k}$. The observed form remains $\eta_{h,k}^* \Pi_{\text{prod}}^* \Pi_{\text{prod}} \eta_{h,k}$, with its original source comparison Gram. The cyclic $\chi_{h,k}$ -potential has its own connection (XD.u1)–(XD.u7); this injection, left inverse and transverse map give the precise comparison with the product potential.

On the complete primary slice $h = (s - \rho)^m$, $t_i = 0$, the exact product period matrix, source-unit character projectors, terminal arithmetic eigenclass and finite-field boundary are proved in (BTP.1)–(BTP.11) and (BTF.1)–(BTF.11). The complex invariant dimension is $(m^k + m(-1)^k)/(m+1)$; the actual finite-field inertia dimension is that number when $kc = 0$ and zero otherwise, with $c = -(-\rho)^{m+1}/(m+1)$. The original unit-weighted coefficient map and all its factorial losses are (PC30a)–(PC30g) and (BPC.1)–(BPC.14), on precisely their stated coefficient-map domains. These full proofs retain the one-factor vanishing as its $k = 1$ case and carry each surviving tensor character to its actual boundary stalk.

Result R69: actual theta dilation and the moving-parameter flow. On the original theta source use $U_a F(x) = F(x/a)$ for positive real a ; its Mellin action is a^s . The two-leg action is $(\phi, \psi) \mapsto (U_a \phi, a U_{1/a} \psi)$. The complete scaling proof computes the original source boundary of its packet section and the finite cocycle, so the quotient section intertwines this action with the full exponential of the original jet operator, including all nilpotents. For the h exponential family the regular connection operator is

$$L = u \partial_t - s, \quad [u \partial_s + h - t, L] = 0, \quad \frac{dC_b}{db} = -C_b A_h(t + ub), \quad C_0 = I.$$

The two nonzero commutators are $-u$ and $+u$. The finite matrix ODE defines the entire transport; on the admitted period domain it is exactly $\Pi_h(u, t)^{-1} \Pi_h(u, t + ub)$. Its special-fibre matrix is $C_b(0, t) = e^{-bA_h(t)}$. At $b = \log p$ this gives the original p^{-A_h} , corresponding to $U_{1/p}$, for every positive real p . The separate global algebraic curve power map $z \mapsto z^p$ has the additional domain $p \in \mathbb{Z}_{>0}$; there its pullback is $dz/z \mapsto p dz/z$ and the divisor depth is multiplied by that integer. The corrected full source states these distinct parameter domains explicitly.

For the retained reflection-stable cyclic polynomial, put $\varepsilon = (-1)^q$, $q = \deg \chi$, and $RP(S) = P(k - S)$. The full reflection proof gives

$$RA_\chi(t)R = kI - A_\chi(\varepsilon t), \quad RC_b^\chi(u, t)R = e^{-kb} C_{-b}^\chi(-\varepsilon u, \varepsilon t).$$

Here the cyclic potential is $\Phi'_\chi = \chi$, with the original endpoint $\Phi_\chi(0) = 0$. Its reflection gauge is $\exp((\Phi_\chi(k) - kt)/u)R$ on $u \neq 0$; the unsquared contour sign remains. The matrix identity extends through $u = 0$ by the entire ODE, rather than by evaluating that singular gauge. The same proof retains the inverse-transpose dual and the original moment-line factor e^{-kb} .

The same finite transport has its complete original u -derivative. Keep $\Omega_{u,h} = -C_h(t)/u^2 + B_h/u$, $\Omega_{t,h} = -A_h(t)/u$, and $t_b = t + ub$. Differentiating the exact matrix formula at fixed b gives

$$\begin{aligned} \partial_u C_b &= -\Omega_{u,h}(u, t) C_b + C_b (\Omega_{u,h}(u, t_b) + b \Omega_{t,h}(u, t_b)), \\ \partial_t C_b &= -\Omega_{t,h}(u, t) C_b + C_b \Omega_{t,h}(u, t_b). \end{aligned}$$

Indeed $\partial_a \Pi^{-1} = -\Omega_{a,h} \Pi^{-1}$; the target derivative is $\Pi(u, t_b)(\Omega_{u,h} + b \Omega_{t,h})(u, t_b)$. This proves the pullback connection under the actual base map $(u, t) \mapsto (u, t + ub)$, including $b \partial_t$. It is the polynomial restriction of the fully proved (APU23) transport. These equations hold on the original nonzero- u contour domain; the same entire finite-matrix ODE supplies the regular transport through $u = 0$. The reflection and inverse-transpose dual retain the stated parameter pullbacks and original moment-line factor; their derivatives follow by these same inverse and chain rules on the unchanged maps.

Result R70: the original Taylor unit is an exact formal gauge. Let ν be the degree-less-than- d polynomial representing the full v_h , so $\gcd(h, \nu) = 1$. The exact rings are $\hat{B} = \mathbb{C}[s][[u, t]]$ and $\hat{B}_\nu = \mathbb{C}[s, \nu^{-1}][[u, t]]$. The coefficientwise pole quotient is $M[[u, t]]$, $M = \mathbb{C}[s, \nu^{-1}]/\mathbb{C}[s]$. Multiplication by h on M has an exact inverse H : Bezout coefficients for h, ν^n prove surjectivity on each principal part and coprimality proves injectivity. With $E_0 = u\partial_s - t$ the inverse on the pole complex is

$$J = \sum_{n \geq 0} (-HE_0)^n H.$$

The displayed order is essential. Each factor HE_0 raises parameter order, making every coefficient a finite sum. The proper-rational section of the quotient gives the explicit contraction and retraction in (UG8)–(UG11). Multiplication by the full ν then gives

$$\nu(u\partial_s + h - t)\nu^{-1} = u\partial_s + h - t - u\nu'/\nu.$$

It commutes with the same $L = u\partial_t - s$. Its endpoint on E_h is exactly multiplication by v_h ; the signed completed external product recovers exactly $\eta_{h,k}$ after the original cyclic inclusion. The complete (UG1)–(UG19) proof includes every tensor homotopy and the inverse-unit residue dual on its original reflection-stable domain.

The formal gauge carries the complete regular form of the singular-parameter operator. Put $f = \Phi_h - ts$. On the original two formal complex terms define

$$H_j = u^2\partial_u - f + (1 - j)u, \quad j = 0, 1, \quad H_1 D = D H_0, \quad D = u\partial_s + h - t.$$

The commutator of $u^2\partial_u$ with D is $u^2\partial_s$, and that of $-f$ with D is $u(h - t)$. Their sum is uD , canceled by the extra u on the degree-zero term. This proves the cochain identity. These operators act coefficientwise on the stated completed rings. After the specified localization in u , they are $u^2\nabla_u^{(j)}$, with the same degree terms as (APU1)–(APU4). Since ν is independent of u, t ,

$$\nu H_j \nu^{-1} = H_j, \quad H_1(\nu D \nu^{-1}) = (\nu D \nu^{-1}) H_0.$$

These are operators on the transported gauge terms; multiplication by f commutes with ν . The polynomial-to-localized inclusion commutes with H_j . Its cohomology isomorphism, proved by the full contraction J , consequently has inverse commuting with the induced H : multiply the intertwining identity by that inverse on both sides. This proves compatibility without assigning strict horizontality to an uncomputed chain retraction. The exact homotopy and full gauge endpoint remain (UG8)–(UG19).

The existing $L = u\partial_t - s$ satisfies $[H_j, L] = uL$, because its two nonzero terms are $u^2\partial_t$ and $-us$. Including the derivative of $1/u$ therefore gives $[H_j/u^2, L/u] = 0$ on the localized complex. At the original formal fibre H_1 is multiplication by $-\Phi_h$, commuting with the entire $[\nu] = v_h$. On the ordered k -fold complex the same calculation gives $u^2\partial_u - \sum_i (\Phi_h - t_i s_i) + (k - j)u$; the original signed tensor gauge and its source endpoint retain their full degree terms and Taylor units.

Result R71: analytic pole residues, their connection and their endpoint. Return to the original h -potential $\Phi'_h = h$ with its specified additive constant. For fixed $u \neq 0$, write $w = e^{(\Phi_h(s) - ts)/u}$ and let $\mathcal{P}_\nu$ be the finite set of distinct zeros of ν , of size r . The exact pole sequence now is

$$0 \longrightarrow M \xrightarrow{u\partial_s + h - t} M \xrightarrow{F \mapsto (\text{Res}_p(wF ds))_p} \mathbb{C}^{\mathcal{P}_\nu} \longrightarrow 0.$$

The highest pole term $c(s - p)^{-n}$ maps to $-unc(s - p)^{-n-1}$, proving injectivity. A zero-residue Laurent series wF has a meromorphic primitive without a logarithm; taking the principal part of that primitive divided by uw proves exactness in the middle. The lift $1/(w(p)(s - p))$ proves surjectivity. The original polynomial cohomology therefore injects into the analytic localization,

with precisely this residue cokernel. A full basis consists of the original d polynomial classes and the simple-pole classes $1/(s-p)$.

Let $P = \text{diag}(p)$ and let B_0 have each column the original constant-polynomial vector e_0 . Their exact connection and transport are

$$L = u\partial_t - \begin{pmatrix} A_h(t) & B_0 \\ 0 & P \end{pmatrix}, \quad C_{\text{ext},b} = \begin{pmatrix} C_b & K_b \\ 0 & e^{-bP} \end{pmatrix}, \quad K_b = - \int_0^b C_c B_0 e^{-(b-c)P} dc.$$

The off-diagonal term follows from $L(1/(s-p)) = -p/(s-p) - 1$ and the displayed ODE. The full routed period determinant is $\det \Pi_{\text{ext}} = \det \Pi_h (2\pi i)^r \prod_p w(p)$, with positive small-loop orientations. Under the actual unit gauge the weight is w/ν ; all periods agree because $(w/\nu)(\nu F) = wF$.

At the original formal endpoint let Q_0 have columns $-[(h(s) - h(p))/(s-p)h(p)]$ in E_h . Then

$$A_h Q_0 = B_0 + Q_0 P, \quad J_0 = [I \ Q_0], \quad \ker J_0 = \{(-Q_0 c, c) : c \in \mathbb{C}^r\}, \quad J_0 C_{\text{ext},b}(0,0) = e^{-bA_h} J_0.$$

Multiplication by the full v_h after J_0 is the gauge endpoint. These are explicit comparison maps between the analytic extension and the original formal fibre. The complete (AP1)–(AP26) proof retains all higher poles, the ordinary dual, the zero-pole case and the exceptional $p=0$ phase; it proves no evaluation of the formal inverse at a nonzero complex u .

The full connection of this analytic extension now includes its u direction, as proved in (APU1)–(APU23). Write $f(s, t) = \Phi_h(s) - ts$, and let Q have columns

$$Q_p(s, t) = \frac{f(s, t) - f(p, t)}{s - p} - \frac{h(s) - t}{d + 1}, \quad p \in \mathcal{P}_\nu.$$

The first polynomial has degree d and leading coefficient $1/(d+1)$. Its division by $h-t$ has that constant quotient; the derivative correction is zero and the displayed remainder has degree less than d . In the original polynomial and simple-pole columns the exact matrix is

$$\Omega_{u,\text{ext}} = \begin{pmatrix} -C_h(t)/u^2 + B_h/u & -Q/u^2 \\ 0 & -\text{diag}(f(p, t))/u^2 \end{pmatrix}, \quad \partial_u \Pi_{\text{ext}} = \Pi_{\text{ext}} \Omega_{u,\text{ext}}.$$

Apply $\nabla_u = \partial_u - f/u^2$ to $1/(s-p)$ and separate $f(p)/(s-p)$ from its complete polynomial divided difference to prove this formula. The original t -matrix remains $-u^{-1} \begin{pmatrix} A_h(t) & B_0 \\ 0 & P \end{pmatrix}$. The full block flatness, including its off-diagonal block, is proved in (APU8)–(APU12) using these literal remainders.

Put $W_{\text{res}} = \text{diag}(e^{f(p,t)/u})$. The weighted-residue row matrix is $\mathcal{R} = [0, W_{\text{res}}]$; the positively oriented loop row matrix is $(2\pi i)\mathcal{R}$. Differentiation gives

$$\partial_u W_{\text{res}} = -\text{diag}(f(p, t))W_{\text{res}}/u^2, \quad \partial_t W_{\text{res}} = -PW_{\text{res}}/u.$$

Thus $\partial_a \mathcal{R} = \mathcal{R} \Omega_{a,\text{ext}}$ for $a = u, t$; both residue and loop maps are horizontal, with their original $2\pi i$ factors retained. The entire transport has the two derivative identities in R69 with the full extension matrices. Equation (APU23) proves the resulting shifted endpoint and residue diagrams, including $b\Omega_{t,\text{ext}}(u, t + ub)$.

Multiplication by the complete ν commutes with the parameter operators and transports the entire frame; its weight is still w/ν . Its inverse-transpose dual and unchanged endpoint J_0 are proved in (APU18)–(APU22). The original theta-source observation is exactly $\sigma_h[\nu]J_0$, retaining the entire unit. When ν is constant, $r=0$ and every pole block, row and additional loop is empty; the same equations reduce to the full original polynomial connection and its unit gauge. These maps retain the analytic family, the specified formal endpoint and the formal coefficient operations of R70.